Detection of Distributed Scatterers in Multitemporal SAR Interferometry: A Comparison Between CAESAR and SqueeSAR Detectors

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***Abstract*— Synthetic aperture radar interferometry (InSAR) with multiple passes is widely used to monitor deformations of the Earth surface. SqueeSAR and Component extrAction and sElection SAR (CAESAR) are two recently proposed solutions for performing an effective multiacquisition multilook-based filtering of interferograms to contrast the effects of phase noise for weak and/or decorrelating scatterers (DSs). Both methods can be used specifically at the detection stage in InSAR processing chains characterized by a two-stage/two-scale processing strategy, to increase the coverage of monitored points. In this work, we compare the detection performances of the SqueeSAR-based and CAESAR-based detectors. We address the evaluation of the performances on simulated data in the classical Gaussian scenario. Results of experiments on data acquired by the TerraSAR-X (TSX) and COSMO-SkyMed (CSK) sensors are also included to discuss the advantages of one method over the other in real scenarios.**

***Index Terms*— Detection, generalized likelihood ratio test (GLRT), persistent scatterers (PSs), synthetic aperture radar (SAR) interferometry (InSAR), SAR tomography.**

I. Introduction

SYNTHETIC aperture radar interferometry (InSAR) with multiple passes is a powerful technology exploited to monitor Earth surface deformations with applications to many fields of natural or anthropogenic hazards.

Persistent scatterer InSAR (PSInSAR) [1], [2] was the pioneering approach capable of demonstrating, for the first time, the effectiveness of this technology. PSInSAR allows tracking, through an appropriate phase modeling, the phase variations on sequences of SAR acquisitions collected at different epochs. It allows measuring deformation with subwavelength accuracy, typically reaching the order of a few mm/year for deformation rates.

Inspired by PSInSAR, many other persistent scatterer interferometry (PSI) techniques have been proposed over the last two decades [3], [4], [5]. Such techniques are sometimes also referred to as advanced or multitemporal (MT) differential InSAR (A-DInSAR or MT-DInSAR) methods [6].

A popular A-DInSAR method is based on a two-scale (two-stage or two-step) processing [7], [8]: a processing at low resolution [3] is, in this case, first carried out to estimate the signal components necessary for calibrating the data for the subsequent (second-stage) high-resolution analysis [7], [8], [9]. More in detail, the first, small-scale (i.e., low-resolution) step analyzes the multipass-DInSAR data stack to separate the atmospheric phase screen (APS), sometimes referred to as atmospheric phase delay component, from possible nonlinear (low-resolution/small-scale) deformation signal. A multilook is, in such a case, exploited to generate the low-resolution interferometric data on a spatial grid (typically with a spacing of a hundred of meters) suitable to describe the atmospheric variations. Data are then compensated with such components to perform, in the second step, a refined estimation of the residual deformation signal at a larger scale, i.e., at full or close to the full resolution. A parametric phase model is usually adopted at this stage to extract and monitor persistent scatterers (PSs), typically associated with strong scattering mechanisms localized on man-made structures in urban areas [10]. Frequently, PSs are also located in areas, including weak and/or distributed and time DSs. The problem of identifying and separating pixels at full resolution containing PS from those severely affected by noise was originally formulated as the problem to threshold a quantity referred to as multitemporal interferometric coherence (MIC). Specifically, the MIC measures the degree of matching (in the complex domain) between the data phase values and the parametric phase signal model [1], [2]. MIC was proven to be a very powerful statistic for the identification of PS at full resolution. Later, the problem was framed in the context of the classic detection theory, largely developed in surveillance radars. The traditional methods exploited in this field were, therefore, been borrowed in the A-DInSAR processing [11].

MIC, which exploits only the phase information of the received signal, was compared in [11] to the single-look (SL)

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generalized likelihood ratio test (GLRT) detector, originated from the radar detection context and framed in the SAR tomography context [11]. The comparison highlighted that, because of the use of the entire complex (amplitude and phase) information, SL-GLRT allows achieving, for a given false alarm probability (P_{fa}), a higher detection probability (P_{d}) compared with the MIC on both simulated data and data acquired from an operative system. Furthermore, both MIC and SL-GLRT were shown to be constant false alarm rate (CFAR) detectors, i.e., able to achieve a P_{fa} independent of the noise power.

The aforementioned SL detectors exploit only the measurements in the pixel under testing (full-resolution detection). They generally work well in built environments, e.g., urban areas and infrastructures. In such a case, the main problem can be induced by the presence of multiple PS interfering (due to the layover phenomenon), in the same pixel [12], [13]. For nonurban areas, the problem to face is the detection of DSs distributed over the resolution cell, i.e., DS. Although embedding useful information about the displacement of the surface where they are located, such scatterers (generally, also weak) behave worse than PS in terms of the amount of phase noise. As a consequence, they cannot be detected by SL detectors tailored to PS. Obviously, the possibility to improve the density of measurement points by detecting DS in rural areas has important consequences for many applications [6], above all for the detection and cinematic characterization of landslides [14]. This need has stimulated the recent development of the research of multilook PSI methods [6], [15].

On the one hand, a detection approach able to exploit the decorrelation characteristics of DS is well formulated in the literature, but is still the subject of ongoing research [16], [17]. The main problem is, in this case, to ensure a CFAR over all the pixels of the scene. On the other hand, multilook detectors have been recently proposed to counteract the loss of detection performances of SL detectors in the presence of weak scatterers. Research developments for the detection of weak scatterer and DS have been achieved mainly because of the recent research advances in the field of multilook MT-DInSAR interferometric filtering, specifically SqueeSAR [15] before and Component extrAction and sElection SAR (CAESAR) [18] later. Both methods are based on a proper processing of the (MT) sample covariance matrix, estimated through the spatial averaging (multilook) of all the interferometric SAR pairs, possibly implemented with an adaptive selection of "similar" pixels (adaptive multilook). SqueeSAR [15] "squeezes" all the elements of the sample covariance matrix to estimate (in a maximum likelihood sense) the filtered phases at the different antennas. This technique has been shown to provide an effective interferometric filtering of DS in rural areas, which is also able to restore the phase consistency of SL measurements, that is, the zero phase summation of triplets of acquisitions. The latter operation is often referred to as phase closure, phase triangulation, or phase linking [19].

CAESAR [18] is an alternative multiacquisition interferometric filtering based on the eigenvalue decomposition (EVD) of the sample covariance matrix [20]: the orthogonal vectors provided by EVD, sorted from the highest to the lowest eigenvalue, allow forming dyadic components able to sequentially approximate, in the Frobenious norm (F-norm) sense, the original covariance matrix. The solution can be equivalently viewed as a principal component analysis (PCA) of the data corresponding to the looks exploited in the estimation of the sample covariance matrix.

CAESAR, similar to SqueeSAR, implements a filtering of interferometric SAR data stacks [18] and restores the phase triangulation property [15], [18], [19], [21], [22]. By extracting the dominant component, i.e., the eigenvector associated with the highest eigenvalue, CAESAR provides, as SqueeSAR, an equivalent PS data vector. Because of the EVD decomposition, CAESAR also makes it possible to separate multiple interfering components, typical of urban areas, directly at the interferogram generation stage [18]. In this case, however, the advantage must be traded off against the loss of spatial resolution intrinsic to the multilook operation and typically undesirable in such a scenario.

With reference to the abovementioned two-scale interferometric processing, both SqueeSAR and CAESAR can be exploited to generate a set of filtered multilook interferograms for the low-resolution analysis (first stage). In such a case, as said, the scale of analysis is set to a value compatible to the APS correlation length, i.e., a few hundreds of meters. CAESAR and SqueeSAR can be, however, also exploited after the data calibration (second stage) to implement a multilook detector aimed at improving the performances for the DS detection. The use in the second stage is investigated in this work: the multilook degree here considered (after data calibration in the second processing stage) is limited to avoid a significant degradation of the spatial resolution.

The CAESAR detector (CAESAR-D) was analytically demonstrated to have a false alarm rate independent of the number of the averaged looks [23]. This peculiarity allows an easy and direct implementation of an adaptive multilook strategy in which the number of looks can be spatially adjusted to preserve some spatial details (multiresolution detection). CAESAR-D has been also compared in [23] to the multilook GLRT (MGLRT) detector proposed in [24] derived with reference to weak fully correlated scatterers. It was shown that MGLRT achieves better detection performances, but it has the drawback of the dependence of the false alarm probability on the number of looks, thus making the implementation of multiresolution detection scheme with a fixed false alarm rate a more difficult task.

In this work, we extend the analysis carried out in [23] by performing a comparison of CAESAR-D to the PS decision test applied to the SqueeSAR filtered phase signal, hereafter referred to as SqueeSAR detector (SqueeSAR-D) [15]. In particular, we address the issue of evaluating the CFAR detection property for SqueeSAR-D and comparing its detection performances with CAESAR-D. The latter has been carried out on simulated data, in a Gaussian scenario, by extending the analysis also to the MGLRT detector, which represents the benchmark. CAESAR-D and SqueeSAR-D are also compared, in a multiresolution detection framework, with data acquired by operative systems, specifically TerraSAR-X (TSX) and COSMO-SkyMed (CSK).

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The last remark is dedicated to frame the contribution of this work with respect to the available literature, specifically that addressing the comparison between CAESAR and SqueeSAR. There are several papers comparing the two approaches also with other solutions [22], [25], [26], [27]. All these papers deal with a comparison with reference to the phase filtering. It was shown that, in general, SqueeSAR outperforms EVD (i.e., CAESAR), assuming the knowledge of the true covariance/correlation matrix. This assumption can, however, rarely be done when processing real data. An opposite behavior in terms of measurement point density on real data has been highlighted in [22]. None of these papers investigated a comparisons in terms of DS detection performances for a given false alarm probability. In this work, we perform such comparison in continuity with the analysis carried out in [23] involving CAESAR-D and MGLRT for the detection of weak and fully correlated scatterers.

II. Single Scatterers Analysis: From the Persistent to the Distributed Scatterer

The single scatterers analysis carried out by PSI techniques, including tomographic-based approaches, aims to detect up to one scatterer and estimate a set of geophysical parameters of interest within each range–azimuth pixel of the processed data stack. The geophysical parameters of interest, according to the dimensionality of the analysis, are (3-D case) the elevation related to the residual topography (RT), along with (4-D case) the deformation mean velocity (DMV) and (5-D case) possible seasonal/thermal deformation coefficient [28].

A. Data Model

Let us refer to a stack of N SAR images, all assumed to be range and azimuth focused, coregistered with respect to a given (master) image, and calibrated with respect to the APD and possible small-scale nonlinear deformation [8]. For a given range–azimuth pixel, the single scatterers analysis accounts for the presence of a scatterer by modeling the N-dimensional data vector $\boldsymbol{x}$ as follows [23]:

$$\boldsymbol{x} = \boldsymbol{\gamma} \odot \boldsymbol{a}(\boldsymbol{p}) + \boldsymbol{w} \tag{1}$$

where $\boldsymbol{\gamma}$ and $\boldsymbol{w}$ are the vectors collecting the samples of the scatterer's backscattering coefficient and noise contribution, respectively, at the different antennas; $\odot$ is the Hadamard (elementwise) product for vectors and matrices; and $\boldsymbol{p}$ is the scatterer's *parameter vector*, collecting the geophysical parameters of interest of the PS phase model. The vector $\boldsymbol{a}(\boldsymbol{p})$ is the unit norm (i.e., $\|\boldsymbol{a}(\boldsymbol{p})\| = 1$) steering vector, whose nth component is given by

$$a_n(\boldsymbol{p}) = \{\boldsymbol{a}(\boldsymbol{p})\}_n = \frac{1}{\sqrt{N}} \exp\left(-j2\pi \boldsymbol{\zeta}_n^T \boldsymbol{p}\right) \tag{2}$$

where $(\cdot)^T$ is the transpose operator and $\boldsymbol{\zeta}_n$ is the vector collecting the frequencies (Fourier mate variables) associated with the elements of $\boldsymbol{p}$. Such frequencies depend on the adopted system parameters, such as the transmitted wavelength, the spatial baseline distribution, the acquisition epochs, possibly the mean temperature distribution over the different acquisitions, and so on (see [8], [29]).

From a statistical point of view, $\boldsymbol{\gamma}$ and $\boldsymbol{w}$ are assumed to be distributed as uncorrelated complex zero-mean circular Gaussian random vectors. The covariance matrix of the data, which is also complex zero-mean circular Gaussian random vector as a consequence of the previous assumption, can be written as follows:

$$\boldsymbol{C} = \boldsymbol{a}(\boldsymbol{p})\boldsymbol{a}^H(\boldsymbol{p}) \odot (\sigma_\gamma^2 \boldsymbol{M} + \sigma_{\boldsymbol{w}}^2 \boldsymbol{I}_N) \tag{3}$$

where $(\cdot)^H$ is the conjugate and transpose operator; $\boldsymbol{M}$ is the *structure matrix* of the scatterer, with all ones diagonal elements, accounting for the possible degree of decorrelation, which may affect the scattering [16], [30]; $\boldsymbol{I}_N$ is the $N \times N$ identity matrix; and $\sigma_{\boldsymbol{\gamma}}^2$ and $\sigma_{\boldsymbol{w}}^2$ are the variances of each element of $\boldsymbol{\gamma}$ and $\boldsymbol{w}$, respectively, related to the signal-to-noise ratio (SNR) defined as follows:

$$\mathrm{SNR} = \frac{\sigma_{\boldsymbol{\gamma}}^2}{\sigma_{\boldsymbol{w}}^2}. \tag{4}$$

The covariance matrix in (3) can also be rewritten as follows:

$$\boldsymbol{C} = (\sigma_\gamma^2 + \sigma_w^2)\boldsymbol{a}(\boldsymbol{p})\boldsymbol{a}^H(\boldsymbol{p}) \odot \boldsymbol{\Gamma} \tag{5}$$

where

$$\boldsymbol{\Gamma} = \left(\frac{\mathrm{SNR}}{\mathrm{SNR}+1}\boldsymbol{M} + \frac{1}{\mathrm{SNR}+1}\boldsymbol{I}_N\right) \tag{6}$$

is sometime referred to as *coherency matrix*. Similar to the structure matrix $\boldsymbol{M}$, the coherency matrix $\boldsymbol{\Gamma}$ has all the diagonal elements equal to one. Moreover, it accounts not only for the possible degree of decorrelation affecting the scattering, but also for the SNR of the illuminated scene.

Some considerations about the characteristics of a detectable scatterer and their relationship with matrices $\boldsymbol{M}$ and $\boldsymbol{\Gamma}$ are now in order. The nature of a scatterer is modeled by the off-diagonal elements of the structure matrix $\boldsymbol{M}$, which are close to one for a PS and lower than one (even significantly and generally decreasing with spatial baseline and/or temporal separation) for a DS. The detection capability of classical full resolution detection schemes tailored to PS is appreciable in detecting strong PS, i.e., PS characterized by high values of SNR [11] but drastically decreases in the presence of weak PS (low SNR values). As a consequence, the detectability of a PS is related to the coherency matrix $\boldsymbol{\Gamma}$ in (6), which differently from $\boldsymbol{M}$ accounts for the SNR. A PS is, thus, likely or unlikely detectable by analyzing how close are the off-diagonal elements of $\boldsymbol{\Gamma}$ to one or zero, respectively.

As for the DS, they are typically associated with homogeneous backscattering non-necessarily related to man-made structures (e.g., roads, bare soil, or, in the limiting case, sparse vegetation). Their scattering characteristics are affected by the occurrence of decorrelation from image to image. For this kind of scatterer, the reduction of the off-diagonal elements of the structure matrix $\boldsymbol{M}$ generates also a reduction of those of the coherency matrix Γ. As a consequence, a weak PS may be considered, better than the strong PS, as an approximation of a DS in terms of coherency matrix. Although important for increasing the coverage of deformation measurements, DS is

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difficult to be detected, similar to weak PS, by classical full resolution (SL) PS detectors. Detection algorithms specifically tailored to the decorrelation characteristics of the DS can be designed at the expense of a significant increase of the complexity of the detector [16]; in this article, we rather resort to the approximation of a DS with a weak PS. In other words, we approximate the correlation reduction of a DS interesting the off-diagonal elements of $\boldsymbol{M}$ with a reduction of the SNR of a PS. Accordingly, in the framework of the comparison of the different multilook detectors, in the following, we treat a weak PS mechanism (spatially independent over the exploited looks as modeled in [24]) in analogy to a DS.

We start by summarizing the SL PS detection.

B. SL Detection

Tomography-based techniques frame the single scatterers identification within the detection theory; they solve, pixel-by-pixel, the following binary hypotheses test:

$$\begin{aligned}\mathcal{H}_0 &: \boldsymbol{x} = \boldsymbol{w} \\ \mathcal{H}_1 &: \boldsymbol{x} = \boldsymbol{\gamma} \odot \boldsymbol{a}(\boldsymbol{p}) + \boldsymbol{w}\end{aligned} \tag{7}$$

in order to discriminate between the absence (hypothesis $\mathcal{H}_0$) and the presence (hypothesis $\mathcal{H}_1$) of a scatterer with a fixed level of false alarm.

With reference to a PS, i.e., to a all-ones structure matrix, it can be demonstrated that the GLRT solution of (7) is given by [11], [24]

$$\frac{|\boldsymbol{x}^H \boldsymbol{a}(\widehat{\boldsymbol{p}})|^2}{\|\boldsymbol{x}\|^2} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{8}$$

where

$$\widehat{\boldsymbol{p}} = \arg\max_{\boldsymbol{\xi}} |\boldsymbol{x}^H \boldsymbol{a}(\boldsymbol{\xi})| \tag{9}$$

is the maximum likelihood estimate (MLE) [11] of the parameter vector $\boldsymbol{p}$ and T is the detection threshold, belonging to the interval $[0, 1]$ and set according to the desired level of false alarm.

The detection scheme in (8) and (9) has the CFAR property with respect to the noise, which makes it possible to exploit, for a given level of false alarm, the same detection threshold for each analyzed pixel, regardless of the power of the noise that affects the associated signal. The decision rule in (8) and (9), usually referred to as SL GLRT, was designed to work at the highest spatial resolution achievable with the imaging SAR system. Furthermore, a comparison in terms of detection capability (for a given level of false alarm) and estimation accuracy between the SL-GLRT and the detector based on the MIC statistic exploited in the context of the PS interferometry [1], [2] has been carried out still in [11]. The MIC detector implements the same decision rule in (8) and (9), provided that the data vector $\boldsymbol{x}$ is substituted with the only-phase vector $\boldsymbol{x}_\varphi$, achieved by dropping the amplitude information from each element of $\boldsymbol{x}$. The comparison showed that GLRT outperforms MIC, achieving a higher number of detected PS and more accurate estimation of their parameters of interest (i.e., the vector $\boldsymbol{p}$).

As already mentioned, the SL detector allows detecting PS typically associated with man-made objects. Moving to the detection of DS, the single scatterers analysis should account for the coherence degree of the targets by properly modeling the structure matrix in (3) and, in the meantime, increase the SNR by filtering out the uncorrelated noise. An appropriate modeling of the structure matrix may not be a simple task. A first attempt to derive a scheme tailored to detection of partially coherent scatterers was carried out in [16]. The detector is, in this case, designed by accounting for simple parametric laws of the spatial and temporal decorrelation. However, the dependence of such laws on the unknown decorrelation parameters makes the scheme derived in [16] rather unpractical for computational time reasons, if the same P_{fa} has to be guaranteed in all pixels for any decorrelation law.

C. Multilook Detection

Several approaches can be devised to detect distributed scatterers, typically all based on the use of a multilook strategy.

In [24], the scatterer is modeled as a weak PS, i.e., still fully correlated along the antennas, but characterized by low SNR values, thus approximating the behavior of a DS. The multilook strategy is, in this case, intended to counteract the effect of high levels of uncorrelated noise. More specifically, a set of L pixels (looks), including the one under test, is exploited to implement the following MGLRT rule:

$$\frac{\text{tr}\left[\widehat{\boldsymbol{C}}\boldsymbol{a}(\widehat{\boldsymbol{p}})\boldsymbol{a}^H(\widehat{\boldsymbol{p}})\right]}{\text{tr}\left(\widehat{\boldsymbol{C}}\right)} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} T \tag{10}$$

which is derived by assuming the looks to be uncorrelated and identically distributed (homogeneous) with a Gaussian distribution. In (10), $\text{tr}(\cdot)$ is the trace operator for square matrix; moreover

$$\widehat{\boldsymbol{C}} = \frac{1}{L}\sum_{l=1}^{L} \boldsymbol{x}_l \boldsymbol{x}_l^H \tag{11}$$

is the multilook sample covariance matrix of the looks $\boldsymbol{x}_l$, and

$$\widehat{\boldsymbol{p}} = \arg\max_{\boldsymbol{\xi}} \text{tr}\left[\widehat{\boldsymbol{C}}\boldsymbol{a}(\boldsymbol{\xi})\boldsymbol{a}^H(\boldsymbol{\xi})\right] \tag{12}$$

is the multilook MLE [24] of the parameter vector $\boldsymbol{p}$.

It can be easily verified that for $L = 1$, the detection rule (10)–(12) simplifies to the one in (8) and (9).

Since the trace of the Hermitian product[1] between two (equal size) matrices defines the F-norm inner product, i.e., the scalar product induced by the Frobenius norm [31], and $\boldsymbol{a}$ is a normalized vector, i.e.,

$$\text{tr}\left[\boldsymbol{a}(\widehat{\boldsymbol{p}})\boldsymbol{a}^H(\widehat{\boldsymbol{p}})\right] = 1 \tag{13}$$

it can be noted that the left-hand side in (10) represents the maximum normalized matching between the data-based sample covariance matrix $\widehat{\boldsymbol{C}}$ and the model-based scatterer covariance matrix $\boldsymbol{a}(\boldsymbol{\xi})\boldsymbol{a}^H(\boldsymbol{\xi}) \odot \boldsymbol{M}$ when $\boldsymbol{M}$ is all ones, i.e., corresponding to a fully correlated scattering.

[1]The Hermitian product between two matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ is defined as $\boldsymbol{A}^H\boldsymbol{B}$.

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It is interesting to note that, in the presence of a DS, the detector in (10) selects the (equivalent) PS with the highest match to the actual degree of decorrelation "embedded" in the measured covariance matrix $\widehat{\boldsymbol{C}}$. Also, interesting is to note that, this advantage is achieved over the SL GLRT detector, which uses a highly biased estimate of the covariance matrix. The disadvantage of MGLRT is, however, that the detection threshold in (10) that guarantees a given level of P_{fa} results to be dependent on the number L of averaged looks [23].

Other strategies for the representation of a DS as an equivalent weak PS-based and based on the use of multilook, specifically the sample covariance matrix in (11), have been, however, recently proposed in the literature [15], [32]. They can be used in the detection framework as alternatives to the MGLRT detection. The rationale is to exploit the sample covariance matrix to extract a new data vector associated with a PS (usually referred to as "equivalent PS"), that better approximates, according to an appropriate criterion, the possible DS present in the analyzed pixel. Two different criteria have been proposed in the literature: one in the interferometric context and the other in the tomographic context.

The tomographic approach is based on the use of the CAESAR algorithm [32], which exploits the PCA for extracting the dominant components (in principle, more than one) of the sample covariance matrix. More specifically, by resorting to the EVD, the covariance matrix in (11) can be decomposed as follows:

$$\widehat{\boldsymbol{C}} = \sum_{n=1}^{N} \lambda_n \boldsymbol{u}_n \boldsymbol{u}_n^H \tag{14}$$

where λ_n is the nth eigenvalue and $\boldsymbol{u}_n$ the corresponding eigenvector. Assuming that the eigenvalues are sorted in a decreasing order, i.e., $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$, the dominant component representing the equivalent PS is given by

$$\boldsymbol{x}_{\text{cs}} = \lambda_1 \boldsymbol{u}_1 \tag{15}$$

which is, therefore, the new amplitude and phase data vector subsequently processed for the detection of the DS.

The SqueeSAR approach [15] exploits the sample covariance matrix to perform a maximum likelihood-based estimate of the phase vector corresponding to the equivalent PS [19]. SqueeSAR starts from the following more general expression of the "true" covariance matrix in (5):

$$\boldsymbol{C}(\boldsymbol{\theta}) = (\sigma_{\boldsymbol{\gamma}}^2 + \sigma_{\boldsymbol{w}}^2)\boldsymbol{d}(\boldsymbol{\theta})\boldsymbol{d}^H(\boldsymbol{\theta}) \odot \boldsymbol{\Gamma} \tag{16}$$

where $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_N]^T$ is the vector collecting the phases of the complex (unit amplitudes) vector $\boldsymbol{d}(\boldsymbol{\theta})$, i.e., $d_n = \exp(j\theta_n)$, and $\boldsymbol{\Gamma}$ is the real and nonnegative coherency matrix in (6), which accounts for both the decorrelation and noise effects. According to the Gaussian distribution of the homogeneous and independent exploited looks, the MLE of the vector $\boldsymbol{\theta}$ is given by

$$\begin{aligned}\widehat{\boldsymbol{\theta}} &= \arg\min_{\boldsymbol{\xi}} \operatorname{tr}\left[\widehat{\boldsymbol{C}}\boldsymbol{C}^{-1}(\boldsymbol{\xi})\right] \\ &= \arg\min_{\boldsymbol{\xi}} \operatorname{tr}\left\{\widehat{\boldsymbol{C}}\left[\boldsymbol{d}(\boldsymbol{\xi})\boldsymbol{d}^H(\boldsymbol{\xi}) \odot \boldsymbol{\Gamma}^{-1}\right]\right\}.\end{aligned} \tag{17}$$

However, being $\boldsymbol{\Gamma}$ an unknown matrix, the SqueeSAR algorithm implements the optimization in (17) by means of the substitution

$$\boldsymbol{\Gamma} \leftarrow \widehat{\boldsymbol{\Gamma}} = \operatorname{diag}(\widehat{\boldsymbol{\sigma}})^{-1}\operatorname{ABS}\left(\widehat{\boldsymbol{C}}\right)\operatorname{diag}(\widehat{\boldsymbol{\sigma}})^{-1} \tag{18}$$

where $\operatorname{ABS}(\cdot)$ is the elementwise modulus operator and $\operatorname{diag}(\widehat{\boldsymbol{\sigma}})$ is the diagonal matrix whose diagonal elements are given by the entries of the vector

$$\widehat{\boldsymbol{\sigma}} = \left[\sqrt{\widehat{c}_{1,1}}, \ldots, \sqrt{\widehat{c}_{N,N}}\right]^T$$

$\widehat{c}_{i,i}$ being the (i,i)th entry of $\widehat{\boldsymbol{C}}$. The equivalent PS is given by

$$\boldsymbol{x}_{\text{sq}} = \boldsymbol{d}(\widehat{\boldsymbol{\theta}}) \tag{19}$$

which represents the new phase-only data vector subsequently exploited for the DS detection.

CAESAR [32] and SqueeSAR [15] have been both proposed to perform an advanced filtering of the interferometric stack typically adopted in the MT-DInSAR techniques. CAESAR has been also proposed to perform the DS detection, or better the weak PS detection, by processing the new data vector in (15) with the detection rule in (8), in this case, referred to as CAESAR-D [23]. Similar to CAESAR-D, a SqueeSAR-based detection rule, hereafter referred to as SqueeSAR-D, can be considered by still exploiting the detection rule in (8) for processing the new data vector $\boldsymbol{x}_{\text{sq}}$.

Both CAESAR-D and SqueeSAR-D process an equivalent PS extracted from the available looks: the former by considering the most powerful (principal) component of the data (which, according to the PCA, is extracted from the sample covariance matrix); the latter by performing the MLE of the phase vector of such an equivalent PS (which, according to (17), involves also the sample covariance matrix). Both detectors, therefore, exploit, *by design*, the sample covariance matrix. However, differently from CAESAR-D, SqueeSAR-D needs also the true covariance (or coherence) matrix; see again (17). Substituting the true matrix with an estimate as in (18) obviously leads to a loss in the estimation capability of the algorithm, especially if the sample covariance matrix is exploited, which can be a biased estimate of the true one. In [33], an analysis of the SqueeSAR performances achieved by using a less biased estimate of the true covariance matrix has been carried out. This aspect is, however, out of the scope of this work. In the following, we, therefore, exploit, similar to the original SqueeSAR implementation [15], the sample covariance matrix for the approximation in (18). Bias due to outliers may, however, impact differently the two multilook detectors; see Section IV.

Some considerations are now in order. As for the tomographic approach, it is worth underlining that the amplitude and phase data vector extracted from the Gaussian distributed looks by means of the CAESAR algorithm are not anymore Gaussian distributed: accordingly, CAESAR-D is not designed by following the GLRT approach. However, as demonstrated in [23], in the limiting case of $L = 1$ (SL case), it results $\boldsymbol{x}_{\text{cs}} = \boldsymbol{x}$, and thus, CAESAR-D becomes coincident with the SL GLRT detector.

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Moving to the SqueeSAR-based approach, it is worth noting that the matrix $\widehat{\boldsymbol{\Gamma}}$, which substitutes $\boldsymbol{\Gamma}$ in (17), is usually rank-deficient because of the limited number (L) of looks exploited to build the sample covariance matrix: in this case, the MLE of $\boldsymbol{\theta}$ is achieved by resorting to the generalized inverse (pseudoinverse) $\widehat{\boldsymbol{\Gamma}}^{\dagger}$ of $\widehat{\boldsymbol{\Gamma}}$. However, when $L = 1$ (SL case), it can be shown that $\boldsymbol{x}_{\text{sq}} = \boldsymbol{x}_{\varphi}$; see Appendix A: as a result, in the SL case, SqueeSAR-D becomes coincident with the MIC detector.

Each of the described multilook detection schemes, i.e., MGLRT, CAESAR-D, and SqueeSAR-D, has its own pros and cons. An in-depth comparison between the MGLRT and CAESAR-D has been already carried out in [23]. On the one hand, for a fixed level of false alarm, the detection capability of CAESAR-D results to be slightly worse than that achieved by MGLRT. This is not a surprising result considering that, differently from MGLRT, the CAESAR-D rule exploits only the dominant component (according to the PCA analysis) of the sample covariance matrix. On the other hand, MGLRT guarantees the desired level of false alarm by using a detection threshold, which is dependent on the number of the exploited looks: this is an operative disadvantage for a pixel-dependent multilook strategy (multiresolution processing) typically exploited to account for the heterogeneity of the features characterizing the imaged scenes [23]. On the contrary, for a fixed level of false alarm, CAESAR-D requires the same detection threshold independent of the number of looks; moreover, such a threshold is equal to that required (for the same level of false alarm) by the SL GLRT. All these considerations lead to the conclusion that CAESAR-D provides a good trade-off between detection capability and computational effort, thus making it the most powerful tomographic-based tool for the monitoring of DSs.

As for SqueeSAR, the literature has shown that it is a powerful tool for the interferometric analysis of DS. Cao et al. [22] and Ansari et al. [27] address the comparison of SqueeSAR with CAESAR for the aim of interferometric data filtering and phase linking. Based on the knowledge of the authors of this work, a performances assessment of SqueeSAR in terms of detection capabilities and the investigation of the peculiarities pertinent to the detection framework, such as the dependence of the detection threshold on the number of looks, have been never carried out. The same comment applies to the comparison with the CAESAR-D detection performances. The remaining part of this work is devoted to overcome this literature gap.

III. Performances Assessment

This section addresses a comparative study of the performances achieved by SqueeSAR-D and CAESAR-D schemes for the detection of single scatterers carried out in the elevation RT (s) and DMV (v) parameter space (4-D case). The analysis is conducted in terms of false alarm probability P_{fa} and probability of detection P_{d},[2] evaluated by resorting to Monte Carlo techniques over 10^6 (for P_{fa}) and 10^4 (for P_{d}) realizations of simulated data. The data have been generated according to the model specified in Section II and by exploiting the system parameters of a TSX stack of $N = 32$ images, characterized by a span of spatial and temporal baselines of 432 m and 528 days, leading to the s and v Rayleigh resolutions of 23 m and 1 cm/year, respectively. Accordingly, the RT interval $[-60, 60]$ m and the DMV interval $[-2.5, 2.5]$ cm/year have been uniformly sampled with a spacing of 3 m and 0.1 cm/year, respectively. The detectors have been applied to the simulated data with two different values of L, i.e., $L = 4$ and $L = 9$, which are the number of looks forming the 2×2 and 3×3 moving average window, respectively. The phase estimation performed by SqueeSAR has been achieved by resorting to a modification of the iterative implementation suggested in [19]; see Appendix B.

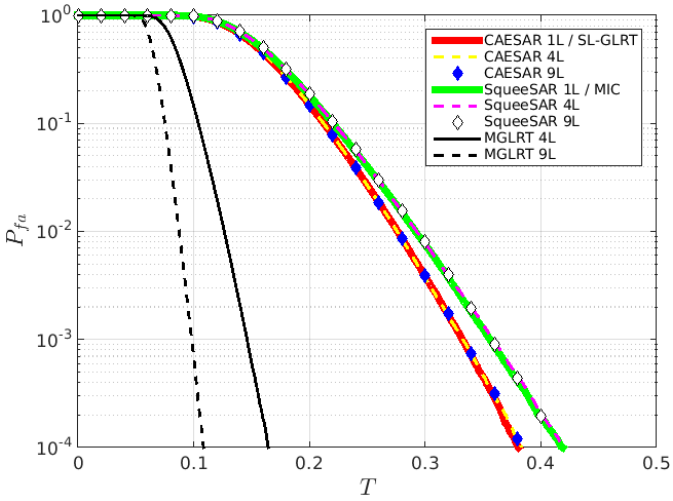


Fig. 1. P_{fa} versus detection threshold T for the MIC, SL-GLRT, MGLRT, CAESAR-D, and SqueeSAR-D detectors. The multilook implementations have been conducted for $L = 9$ and $L = 25$ looks.

In Fig. 1, the P_{fa} is plotted versus the threshold T for the exploited number N of images and L of looks. Along with the curves of SqueeSAR-D and CAESAR-D, those relevant to the well-established detectors specifically, MIC (SL SqueeSAR-D) and SL-GLRT (SL CAESAR-D) schemes in (8) as well as the MGLRT scheme in (10) are also plotted for comparison purposes.

As expected, all the detectors have a P_{fa} decreasing with T. Moreover, differently from the MGLRT for which, fixed the P_{fa}, the threshold T is a decreasing function of L, CAESAR-D and SqueeSAR-D result to have a P_{fa} invariant with the number of looks and, therefore, coincident with the one of the SL case, i.e., SL-GLRT and MIC schemes, respectively. This result, was already known for CAESAR-D and analytically demonstrated in [23], has important consequences for the applicability of the detectors to real data in the presence of a variable (i.e., pixel-dependent) number of looks, as shown in [23] and discussed in Section IV. In addition, similar to CAESAR-D and all the discussed detection schemes, SqueeSAR-D ensures the CFAR property with respect to the noise level.

[2]The measures of performances are defined as $P_{\text{fa}} = P(\mathcal{D}_1|\mathcal{H}_0)$ and $P_{\text{d}} = P(\mathcal{D}_1|\mathcal{H}_1)$, where $P(\mathcal{D}_i|\mathcal{H}_j)$ is the probability of deciding for the ith hypothesis when the jth is true.

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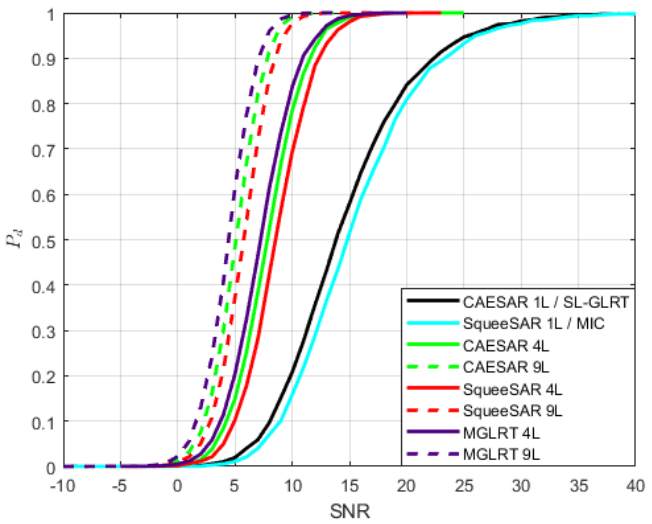


Fig. 2. P_{d} versus SNR [dB] for the MIC, SL-GLRT, MGLRT, CAESAR-D, and SqueeSAR-D detectors, for $P_{\mathrm{fa}} = 10^{-4}$.

Fig. 2 shows the P_{d} of all the analyzed detectors plotted versus the SNR defined as in (4) for the same values of N and L, as in Fig. 1. Detection thresholds are set to the values that guarantee, for both detectors, the same false alarm probability, specifically $P_{\mathrm{fa}} = 10^{-4}$. We note first of all that for a fixed value of SNR, all plots show a gain of P_{d} associated with an increase of L. Particularly evident is the gain in the detection probability of weak PS (say with SNR lower than 15 dB for this acquisition configuration) achieved even by using even only four looks. In can be also noted that the P_{d} of the MGLRT represents an upper bound for the detection performances of both CAESAR-D and SqueeSAR-D. However, CAESAR-D always performs better than SqueeSAR-D, with an increase, which is almost constant with L. In other words, the P_{d} curves gain discussed in [11] of SL-GLRT (i.e., SL CAESAR-D) over MIC (i.e., SL SqueeSAR-D) is retained for all L values.

The dependence of the P_{fa} and P_{d} curves on the number N of images can also be investigated. However, in several references (see [11], [16], [24]), it has been shown that an increase of N produces a reduction of the detection threshold required to achieve a given value of P_{fa} and a consequent increase of P_{d}. This is, substantially, the qualitative behavior of all detector schemes, and then, the dependence of the performances on N will not be further investigated for brevity reasons.

The above analysis has shown two main results in the comparison between CAESAR-D and SqueeSAR-D.

1) Given a value of P_{fa}, the detection threshold of both detectors is independent of the number L of looks.
2) Given a value of SNR, the P_{d} of CAESAR-D always overcomes that of SqueeSAR-D.

As for the point 2, it is worth highlighting that the comparison has been carried out by assuming, according to the model in Section II, fully correlated (although weak) scatterer along the exploited acquisitions. The case of a scatterer imaged at different antennas with a given level of decorrelation is certainly of interest. Detectors able to account for the decorrelation degree can be derived; see [16] for the SL case and [17] for the multilook case. It should be, however, noted that the detection threshold is, in this case, dependent on the degree of correlation, thus leading to a more complicated detection scheme in which the threshold has to be adapted pixel by pixel. This analysis is out of the scope of this work.

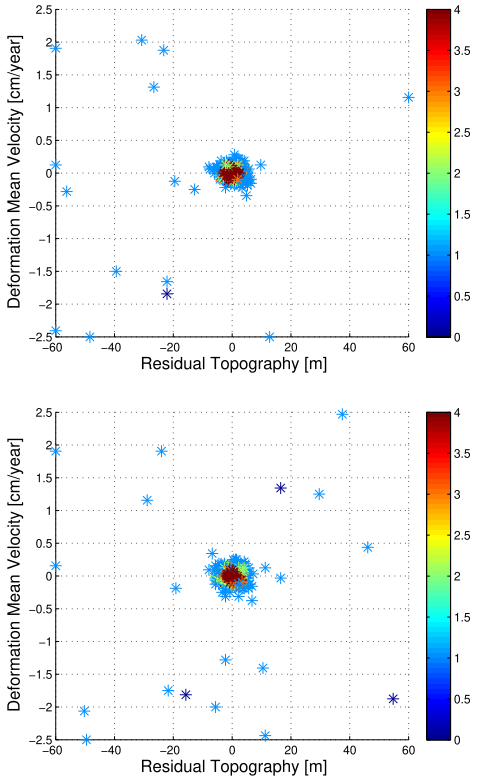


Fig. 3. Scatter plot of the estimated RT and mean deformation velocity. A scatterer with SNR equal to 12 dB has been placed in the zero position. The results of the parameter estimation for CAESAR-D and SqueeSAR-D are shown in the images on the top and on the bottom, respectively.

Finally, a comment on the accuracy of the estimation of the scatterer parameters is in order. The results reported in Fig. 3 representing the scatter plot of the estimated DMV and RT of a nonmoving, zero RT scatterer show a better accuracy of the estimation carried out by CAESAR-D with respect to SqueeSAR-D. These results are fully consistent with those reported in [11], addressing a detailed performance comparison the between SL-GLRT and MIC, i.e., the estimators to which CAESAR-D and SqueeSAR-D reduce for the SL case. More specifically, the minimum square errors achieved by MIC and SL-GLRT, the Cramer–Rao lower bound, and the scatter plots of the estimated parameters for several values of the SNR were there provided.

IV. Experimental Results on Real Data

Performance comparison of CAESAR-D and SqueeSAR-D for the detection of single scatterers has been carried out on SAR datasets acquired by operative SAR systems relevant to a rural area in South Italy and to residential districts in the cities of Bucharest (Romania) and Rome (Italy). For brevity reasons, the results of the Rome test case are only reported numerically, in a table, at the end of the section.

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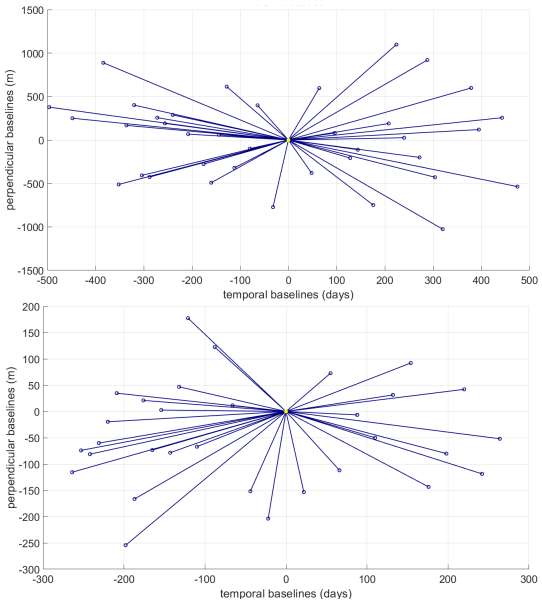


Fig. 4. Spatiotemporal distribution of baselines of the CSK (top) and TSX (bottom) datasets. Acquisitions are represented by circles. The yellow circle in each figure corresponds to the master acquisition.

TABLE I
PARAMETERS OF THE PROCESSED DATASETS

Parameter	COSMO-SkyMed	TerraSAR-X
number of images	38	32
acquisition interval	January 2017 - September 2019	July 2011 - December 2012
slant range resolution	1.44 m	1.25 m
azimuth resolution	2.6 m	3.3 m
elevation resolution	5.49 m	23 m
DMV resolution	0.59 cm/year	1 cm/year
wavelength	3.1 cm	3.1 cm
sensor-scene distance	747.92 km	629.4 km
look angle	34.72°	37.32°
acquisition mode	stripmap	stripmap
test scene size (az × rg)	1000 × 1300 pixels	800 × 1200 pixels
orbit direction	ascending	descending

The first dataset refers to a number of $N = 38$ CSK images acquired in the strip-map mode, with a spatial resolution of about 3 m in both azimuth and ground range directions. The testing case corresponds to the rural area of Sant'Angelo in Formis in the north of Naples, considered also in [23] and including an important Benedictine Basilica located close to a landslide. The scenario is, therefore, the most appropriate for the context of our analysis, being characterized by the presence of distributed/weak scatterers.

The second dataset consists of $N = 32$ TSX images acquired in strip-map mode, with a spatial resolution of again about 3 m in both azimuth and ground range directions. The images are relevant to the residential National Arena area of Bucarest, thus making this test case alternative to the CSK one relative to a rural scenario.



Fig. 5. Temporally averaged amplitude for (top) CSK dataset—Sant'Angelo in Formis area and (bottom) TSX dataset—National Arena area. Azimuth is vertical, and range is horizontal.

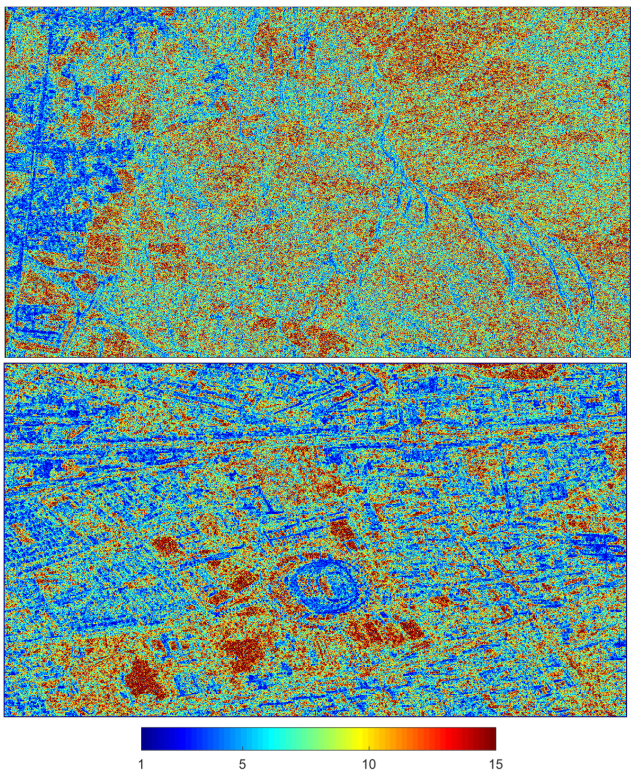


Fig. 6. Spatial distribution of the number of SHP for (top) CSK dataset—Sant'Angelo in Formis area and (bottom) TSX dataset—National Arena area.

The acquisition distributions in the temporal/spatial (orthogonal) baseline domain are shown in Fig. 4 for (top) CSK

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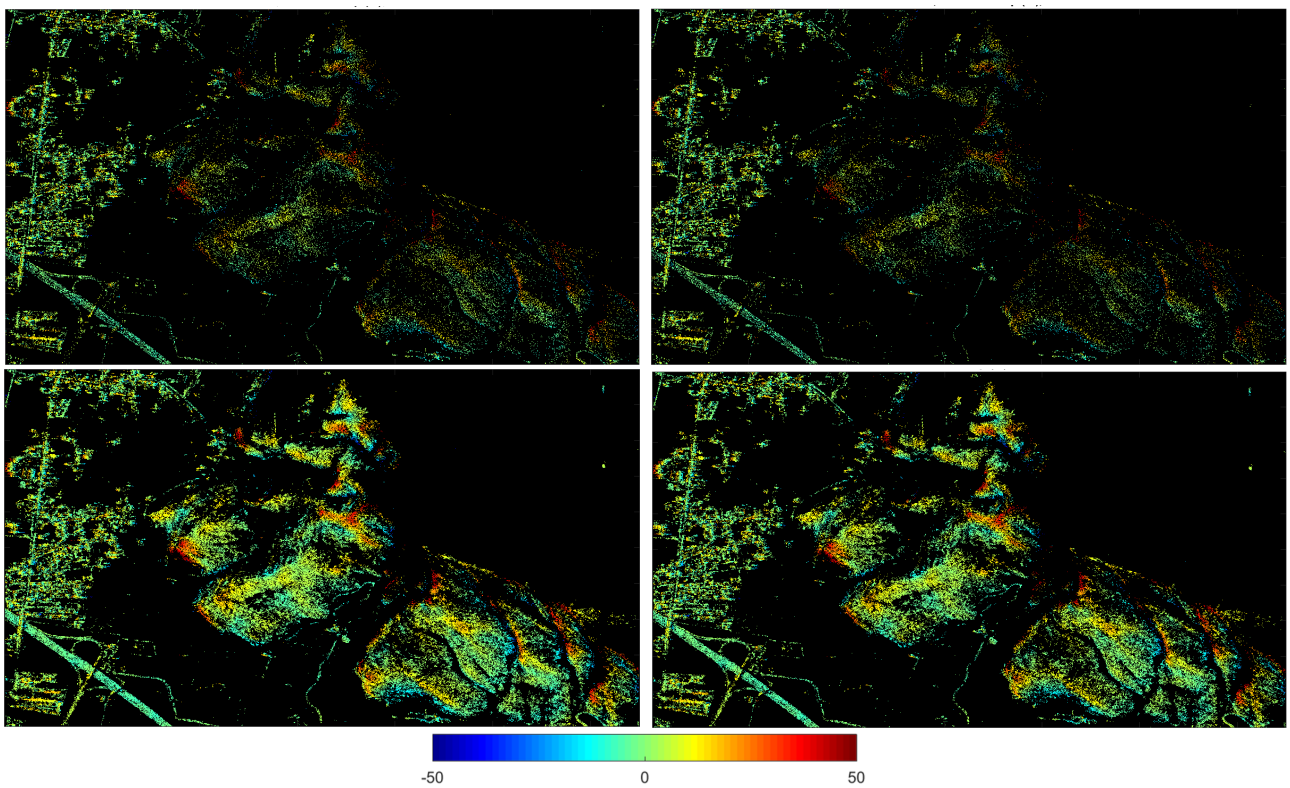


Fig. 7. Spatial distribution of the scatterers detected in the Sant'Angelo in Formis area (CSK dataset) by (left column) CAESAR-D and (right column) SqueeSAR-D in (top line) SL and (bottom line) adaptive multilook cases. Color map is set according to the estimated RT values in meters.

TABLE II
NUMBER OF DETECTED SCATTERERS, SANT'ANGELO IN FORMIS CSK DATASET

CAESAR-D	SqueeSAR-D	
71001	59362	**single-look**
176534	156829	**adaptive multilook**

TABLE III
NUMBER OF DETECTED SCATTERERS, TSX DATASET

CAESAR-D	SqueeSAR-D	
163797	153666	**single-look**
210182	216019	**adaptive multilook**

TABLE IV
NUMBER OF DETECTED SCATTERERS, ROME CSK DATASET

CAESAR-D	SqueeSAR-D	
50010	44555	**single-look**
67623	63838	**adaptive multilook**

and (bottom) TSX dataset: the corresponding span of spatial and temporal baselines results to be of 1462 m and 971 days, and of 432 m and 528 days, respectively. Table I synthesizes the characteristics and parameters of the two datasets.

A distinctive difference between the acquisition features of the two datasets is related to the spatial baseline span, which, for the TSX dataset, is much lower than the CSK case. Another distinctive difference is in the scene scattering characteristics. The temporally averaged amplitude images in Fig. 5 for the cases of (top) CSK and (bottom) TSX show the differences in the typology of the scenario.

For all the analyzed data stacks, the acquisitions were preliminary phase calibrated with respect to the contributions associated with the APS and large-scale deformations.

The adaptive multilook strategy exploited for CAESAR-D and SqueeSAR-D is compliant with a multiresolution analysis. It is based on the identification of a set of statistical homogeneous pixels (SHPs) belonging to a broader search region, usually corresponding to a window centered on the pixel to be processed [15], [23]. The identification of the SHP has been conducted by means of the Kolmogorov–Smirnov (KS) test, as implemented in [23], within a 5×5 search window and a significance level $\alpha = 0.5$. These parameters have been set to identify a number L of looks (SHP) able to trade off the reduction of the noise effect and the preservation of the spatial resolution (see [23, Appendix B] for more details). On the one hand, we tried to preserve the statistic of the pixels corresponding to stable structures (PS) in urban areas, where a low number L of averaged looks is desired. On the other hand, a higher number of SHP shall be selected in distributed texture, usually characterized by a backscattering with low values of SNR and associated with the presence of weak/distributed scatterers. The achieved spatial distributions of the number of SHP selected on the two dataset are shown in Fig. 6 with a scale saturation set to 15 pixels. Obviously, there is a significant difference between the distributions due to the abovementioned variations of the scattering characteristics of the investigated scenario. The Sant'Angelo in Formis dataset has, in fact, larger values of the number of selected SHP and, hence, more significant for the analysis carried out in this work.

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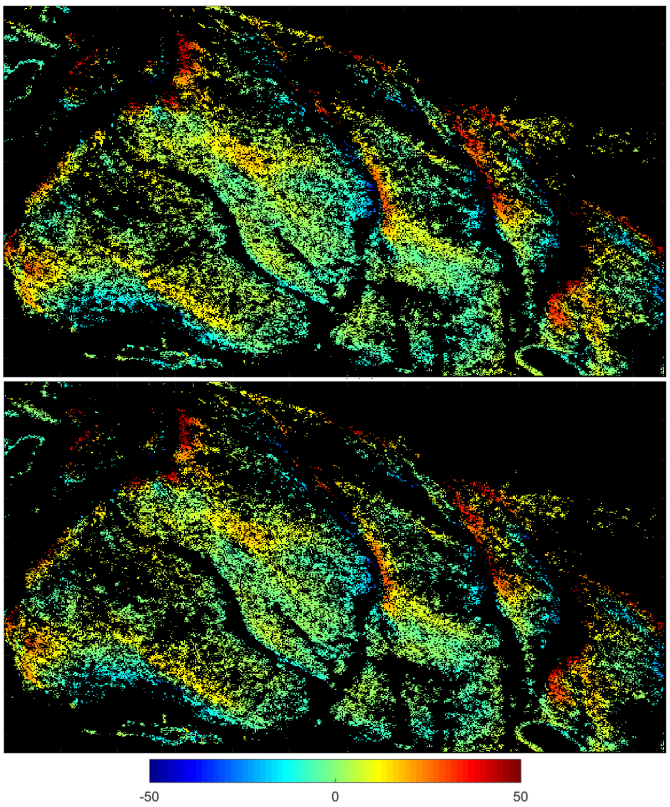


Fig. 8. Zoomed-in view of the bottom-right corner of the CSK results in the bottom row of Fig. 7 - CAESAR-D (top), SqueeSAR-D (bottom).

Multilook detection performances on the real datasets have been evaluated in the RT and DMV space (4-D case) by applying the CAESAR-D and SqueeSAR-D with the adaptive look selection strategy described above. Moreover, the SL detection processing has also been carried out for comparison purposes. The exploited detection threshold guarantees a probability of false alarm of 10^{-4} and has been set by querying, for each detector, the corresponding false alarm probability curve evaluated by resorting to Monte Carlo techniques over 10^6 realizations of data simulated according to the parameters of the processed real dataset.

The CSK dataset was processed by uniformly sampling the RT interval $[-50, 50]$ m and the DMV interval $[-2, 2]$ cm/year with a spacing of 1 m and 0.1 cm/year, thus compatible with the elevation and DMV Rayleigh resolutions in Table I.

The spatial distribution of the scatterers detected in the Sant'Angelo in Formis area is shown in Fig. 7, with the color map representing the estimated values of RT (expressed in meters). The top line is relevant to the SL case, i.e., SL-GLRT (or, equivalently, SL CAESAR-D) on the left and MIC (or, equivalently, SL SqueeSAR-D) on the right. The bottom row shows the adaptive multilook results of CAESAR-D (left) and SqueeSAR-D (right). The comparison of the images in the top and bottom rows highlights the dramatic improvement in measurement points coverage achieved by the multilook in the detection of weak DSs in rural areas.

A zoom-in view of the results in Fig. 7 relevant to the bottom right corner is presented in Fig. 8. CAESAR-D results are in this case shown in the top image, while SqueeSAR-D results are shown in the bottom image. Table II reports the number of detected scatterers. In the SL case, CAESAR-D (or, equivalently, SL-GRLT) performs better than SqueeSAR-D (or equivalently MIC), in agreement to the outcomes of [23]. In the multilook case, CAESAR-D achieves a significant improvement with about 12.6% increase of the detected scatterers with respect to SqueeSAR-D.

The TSX dataset has been processed by setting the span and discretizing the RT and DMV domain, as described in Section III. The spatial distributions of the detected scatterers are shown in Fig. 9: the order of the maps and the exploited colormap are the same as in Fig. 7.

Table III reports the number of the detected scatterers. Consistently with analysis in [11], the SL CAESAR-D detects a higher number of scatterers than SqueeSAR-D. On the other hand, for the multilook case, there is a slight increase, quantifiable in 2.8%, of number of detected scatters of SqueeSAR-D with respect to CAESAR-D. A further investigation about the reason of the difference between the simulation outcomes of Section III and real data result associated with this dataset is provided in Section IV.

To complete the analysis, we report also the table corresponding to the detected scatterers of dataset composed by $N = 29$ CSK images acquired in strip-map mode over Basilica San Paolo area of Rome. The span and sampling of the RT and DVM intervals are $[-60, 60]$ m and the DMV interval $[-2.5, 2.5]$ cm/year with the same spacing of the Sant'Angelo in Formis case. Table IV reports the number of detected scatterers. CAESAR-D achieves an improvement over SqueeSAR-D for both SL and adaptive multilook cases. In both situations, there is an increase of the scatterers detected by CAESAR-D compared with SqueeSAR-D, which for the multilook case is about 5.9%.

V. Comments

The existing literature reports the result of comparisons between SqueeSAR and other multiinterferogram SAR filtering approaches based on the use of multilook, including CAESAR. It has been shown that SqueeSAR allows achieving better performances in terms of phase estimation error; see [27]. As also stressed in Section II, this result is a consequence of the fact that SqueeSAR is a maximum likelihood (hence efficient) estimator of the phases. However, to be an MLE, it requires the knowledge of the "true" data covariance matrix. In the reality, such a matrix is never known and must be estimated from the data, see (18), thus leading to an unavoidable performance degradation in terms of accuracy of the estimated phases.

Monitoring of deformations by means of advanced MT interferometric SAR processing involves the problem of detecting the presence of PS/DS. Within the framework of a two-scale processing methodology, the results on simulated and real data considered in this work aim to compare SqueeSAR-D and CAESAR-D in terms of multilook detection of weak-PS/DS scatterers.

For simulated data tailored to the classical Gaussian scenario, the results reported in Fig. 2 clearly show that, the CAESAR-D feature of using both the amplitude and phase information also at the detection stage makes it possible to achieve a higher detection performance with respect to

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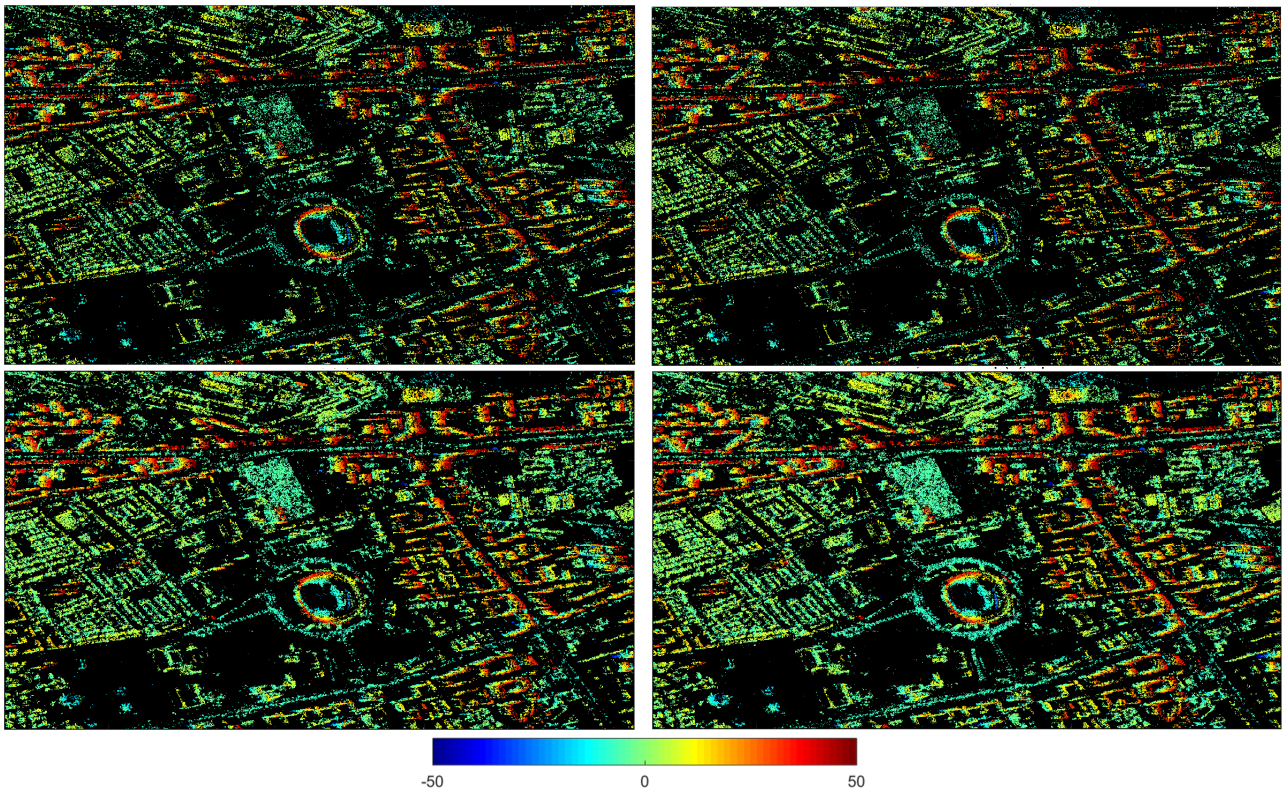


Fig. 9. Spatial distribution of scatterers detected in the National Arena area (TSX dataset) by (left column) CAESAR-D and (right column) SqueeSAR-D in (top line) SL and (bottom line) adaptive multilook cases. Color map is set according to the estimated RT values in meters.

SqueeSAR-D, consistently with the SL analysis in [11]. Essentially, the increase of detection performances of CAESAR-D over SqueeSAR-D is comparable to that of SL case, i.e., to the increase of SL-GRLT over MIC. The gain of CAESAR-D over SqueeSAR is almost independent of the number of looks. Real data can differ to some extent from the fully correlated and weak scattering Gaussian scenario, pertinent to the simulated case. Accordingly, the interpretation of the results can be less simple. We start from the results achieved with the SL processing. In this case, consistently with [11] and simulation results, neglecting the amplitude information (as done by MIC) leads to a decrease of the detection performances on all datasets. On the other hand, for the TSX dataset, the multilook SqueeSAR-D achieved a slight increases in the number of the detected scatterers compared with CAESAR-D. The spatial distribution of the additional detected scatterers, i.e., the scatterers detected by one detector but not by the other, shows that such behavior is mainly associated with the circular pattern located around the stadium; see Fig. 10. The increase of detection achieved by SqueeSAR-D results to be associated with those pixels where the signal is characterized by a lower degree of amplitude homogeneity. This is confirmed by the analysis of the contrast C [34], [35], [36], evaluated pixel-by-pixel and defined as the following function of the amplitude response over the different images:

$$C = \frac{\sqrt{\dfrac{1}{N-1}\displaystyle\sum_{n=1}^{N}\left(|x_n| - \frac{1}{N}\sum_{i=1}^{N}|x_i|\right)^2}}{\dfrac{1}{N}\displaystyle\sum_{i=1}^{N}|x_i|}. \tag{20}$$

The contrast C represents the standard deviation, normalized to the mean, of the amplitude distribution; it is a nonnegative number, which reaches the minimum value for a constant, i.e., fully homogeneous, amplitude distribution (standard deviation equal to zero) and increases with the amplitude distribution variability (inhomogeneity).

Fig. 11 reports the normalized histogram of the contrast on the class of pixels detected by both SqueeSAR-D and CAESAR-D (blue curve) and by SqueeSAR-D only (red curve). It is evident the increase of the contrast for the pixels detected only by SqueeSAR-D, which confirm their higher amplitude inhomogeneity.

It is worth noting that a similar result can be achieved also by analyzing the Rényi Entropy H_{R} [34], given by

$$H_{\mathrm{R}} = \frac{1}{1-\alpha}\log\sum_{n=1}^{N}\left(\frac{|x_n|}{\sum_{i=1}^{N}|x_i|}\right)^{\alpha} \quad \alpha = 2. \tag{21}$$

However, it can be easily verified that the contrast in (20) and the entropy in (21) are monotonically related each other by

$$H_{\mathrm{R}} = -\log\left(\frac{N-1}{N^2}\right) - \log\left(C^2 + \frac{N}{N-1}\right) \tag{22}$$

thus making the H_{R} analysis redundant.

A possibility to counteract the amplitude inhomogeinity is provided by the normalization of the data amplitude over the different looks; see [15]. This is an option that is suggested also for CAESAR filtering in [22]. The choice to avoid normalization in this study is, first of all, motivated by the fact that we prefer to compare the detectors under the Gaussian assumption for the measured data. Moreover, it has been noticed from experimental tests not included here to avoid burdening the

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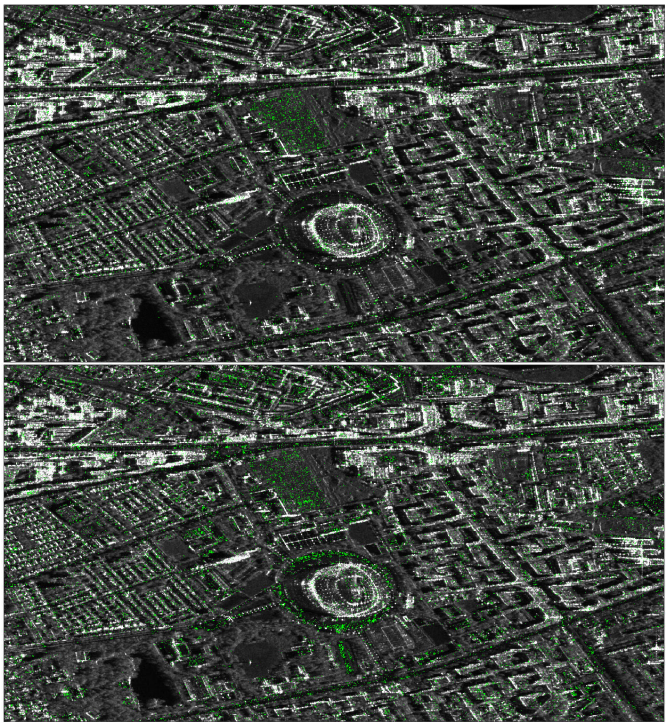


Fig. 10. Spatial distribution (green) of the additional scatterers detected by (top) CAESAR-D relative to SqueeSAR-D and vice versa (bottom), in case of adaptive multilook, TSX dataset.

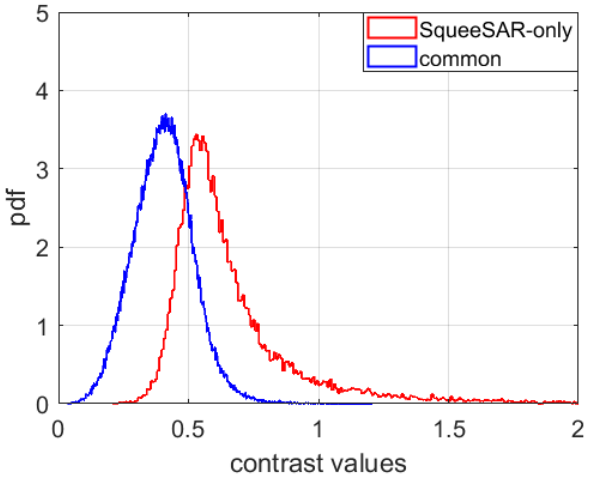


Fig. 11. Comparison of the amplitude's contrast between the points detected by both CAESAR-D and SqueeSAR-D detectors (blue plots) and the points detected only by the later detector (red plots), TSX dataset.

content that the normalization for CAESAR would lead, first of all, to the dependence of the false alarm curves on the number of looks. The normalization would also lead to the divergence between CAESAR-D and SL-GLRT for $L = 1$: in this case, CAESAR-D would rather reduce to the MIC (i.e., SL SqueeSAR-D) for the SL case. Other solutions for a robust covariance estimation are possible; however, a deeper analysis of this aspect is outside the scope of this work.

Conversely, for the CSK dataset's improvement, particularly significant for the rural scenario case of Sant'Angelo in Formis, CAESAR-D overcomes SqueeSAR-D on the detection

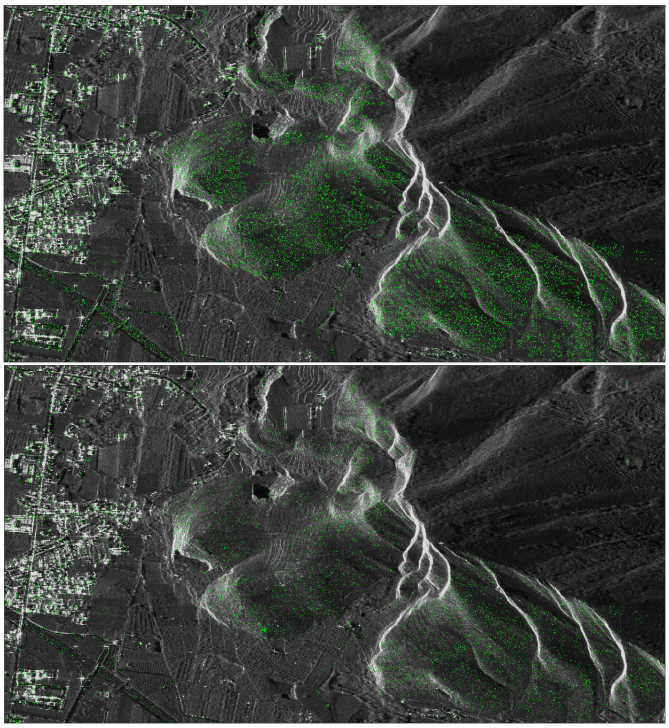


Fig. 12. Spatial distribution (green) of the additional scatterers detected by (top) CAESAR-D relative to SqueeSAR-D and vice versa (bottom), in case of adaptive multilook, CSK dataset.

of weak scatterers on both SL and multilook cases; see Table II and Fig. 12.

VI. Conclusion

The CAESAR-D was compared with the MGLRT in [23] on simulated data and on data acquired by operative systems. MGLRT was shown to achieve better detection performances with respect to CAESAR-D, at the expense of a dependence of the false alarm probability on the number of averaged looks, which represents an issue for the implementation of adaptive multilook strategies on real data.

In this work, we have extended the performance comparison by also considering the SqueeSAR-D. Specifically, on simulated Gaussian data, we have compared CAESAR-D and SqueeSAR-D with respect to the benchmark MGLRT with a fixed false alarm probability. The experiments have shown also that the SqueeSAR-D, similar to the CAESAR-D, shows false alarm curves, which are independent of the number of averaged looks. CAESAR and SqueeSAR-D have been also compared, again a fixed false alarm probability and an adaptive multilook strategy, on real data acquired by CSK and TSX sensors.

Overall, it has been also shown that CAESAR-D allows generally achieving better detection performances with respect to SqueeSAR-D. For the latter, different implementation were tested, and a solution based on a modification of the iterative scheme proposed in [19] has been considered to achieve higher detection performances reasons.

More specifically, the analysis conducted on simulated and real data allows stating that the following hold.

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1) Under the Gaussian scenario, MGLRT and CAESAR-D allow detecting a higher number of scatterers with respect to SqueeSAR-D. Considering that, for SL, MGLRT and CAESAR-D both degenerate in the SL-GLRT, and that SqueeSAR-D reduces to the MIC detector, the achieved results can be considered an extension of the results of [11] where it was shown that SL-GLRT outperforms MIC.
2) CAESAR-D shows detection curves that, compared with those of SqueeSAR-D, are closer to MGLRT, representing the benchmark. Differently from CAESAR-D and SqueeSAR-D, MGLRT, however, suffers for the variability of the false alarm rate on the number of averaged looks; see [23].
3) The feature of SqueeSAR-D to retain only the phase information compared with CAESAR-D that exploits the whole complex (amplitude and phase) information leads, on the one hand, generally, to a decrease of the detection performances also on real data. On the other hand, when the dataset is characterized by higher inhomogeneities of the amplitude response, multilook SqueeSAR-D may overcome for some areas CAESAR-D in terms of detected scatterers.

Further developments on the latter aspects as well as the derivation of optimal detection strategies for DSs are worth of future investigations.

Appendix A

In this appendix, we show that, in the SL case, the MLE $\widehat{\boldsymbol{\theta}}$ provided by (17) is equal to the phase vector of the exploited data vector $\boldsymbol{x}$, i.e., $\widehat{\theta}_n = \angle x_n$, $n = 1, \ldots, N$.

To this aim, after underlining that a positive matrix (vector) is an all-positive entries matrix (vector), we resort to the following [37].

Theorem: Let $\boldsymbol{\Gamma}$ be an $N \times N$ positive matrix. Then, the largest eigenvalue μ of $\boldsymbol{\Gamma}$ is positive, and there is a positive vector $\boldsymbol{v}$, such that $\boldsymbol{\Gamma}\boldsymbol{v} = \mu\boldsymbol{v}$.

The coherency matrix $\boldsymbol{\Gamma}$ is positive. Furthermore, it is positive definite. By sorting, without loss of generality, its eigenvalues μ_n in an ascending order, it is $\mu_1 > \cdots > \mu_N > 0$: according to the above theorem, there is a set $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_N$ of eigenvectors, such that $\boldsymbol{v}_1$, corresponding to the maximum eigenvalue μ_1, is a positive vector.

Let us turn back to the MLE in (17). In the SL case, by exploiting the EVD of the coherency matrix $\boldsymbol{\Gamma}$, it can be rewritten as follows:

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\xi}} \sum_{n=1}^{N} \frac{1}{\mu_n} \left| \boldsymbol{x}^H \left[\boldsymbol{a}(\boldsymbol{\xi}) \odot \boldsymbol{v}_n \right] \right|^2 \tag{23}$$

which is the weighted superposition of the (squared amplitude of the) projections of $\boldsymbol{x}$ along the orthogonal directions $\boldsymbol{a}(\boldsymbol{\xi}) \odot \boldsymbol{v}_n$. The minimum is achieved when the rotation induced by the steering vector $\boldsymbol{a}(\boldsymbol{\xi})$ on the eigenvectors $\boldsymbol{v}_n$ aligns the data $\boldsymbol{x}$ along the less powerful direction, which is determined by $\boldsymbol{v}_1$: being the latter a positive vector, such an alignment occurs when the data vector is in phase with the (phase only) steering vector, i.e., when $\xi_n = \angle x_n$ for each $n = 1, \ldots, N$.

Appendix B

Compared with the implementation of CAESAR-D, which requires essentially the use of available and simple SVD decomposition tools, the implementation of SqueeSAR-D is certainly a more complicated task. It can be approached by using standard nonlinear optimization tools, which are available even in generic signal processing software. Or, alternatively, by resorting to the following iterative implementation introduced in [19]:

$$\theta_n^{(l+1)} = \angle \sum_{\substack{k=1 \\ k \neq n}}^{N} \{\boldsymbol{\Gamma}^{-1}\}_{n,k} e^{j\theta_k^{(l)}} \{\hat{\boldsymbol{C}}\}_{n,k} \tag{24}$$

where the argument in the superscript round brackets is the iteration number and $\{\boldsymbol{A}\}_{n,k}$ is the n, k element of the matrix $\boldsymbol{A}$. Equation (24) shows that SqueeSAR estimates the nth phase value θ_n by a weighted averaging of all the estimates of θ_n coming from the other elements of the sample covariance matrix (i.e., from the multilook interferograms) on the nth row, after proper compensation of the phases of the other images ($\theta_k, k \neq n$) achieved at the previous iteration. The weighting is carried out by the inverse of the (real and nonnegative) coherency matrix. However, according to our experience, the best performances in terms of detection capability on both simulated and real data cases [compared with either standard non linear optimization tools or (24)] have been achieved by using a modification of the iterative algorithm suggested in [19]. In particular, we considered the following iterative scheme:

$$\theta_n^{(l+1)} = \angle \sum_{\substack{k=1 \\ k \neq n}}^{N} \{\mathrm{ABS}(\boldsymbol{\Gamma}^{-1})\}_{n,k} e^{j\theta_k^{(l)}} \{\hat{\boldsymbol{C}}\}_{n,k}. \tag{25}$$

The absolute value to the (real) element of the inverse matrix is the modification carried out on the algorithm suggested in [19] and prevents the introduction of π phase jumps associated with possible negative elements. Results not reported here for brevity showed that the implementation with the iterative formula in (24) achieves better detection performances on simulated and real data with respect to both the original solution proposed in [19] and the fminuc MATLAB optimization tool.

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